What Variables Affect Out-of-Distribution Generalization in Pretrained Models?

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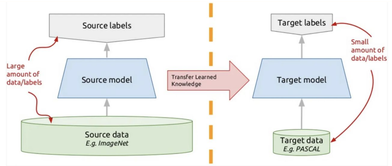
Motivation

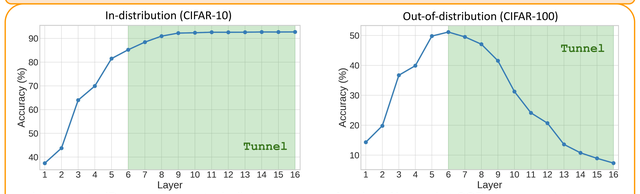


Research Question: How do in-distribution (ID) training data properties (resolution, # samples, and # classes), data augmentation, and architectural choices impact out-of-distribution (OOD) generalization and DNN representations?

Research Question: Why do methods validated on low-resolution datasets with a small number of classes (e.g., CIFAR-10) fail to generalize to high-resolution datasets with a large number of classes like ImageNet-1K?

In a highly controlled study, we examine these questions through the lens of the Tunnel Effect Hypothesis, which is closely related to intermediate Neural Collapse

The Tunnel Effect Hypothesis

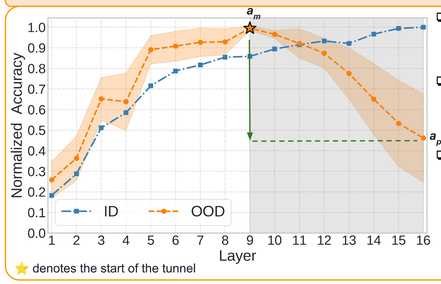


- Linear probe ID accuracy monotonically increases as a function of layers, but OOD accuracy only increases until the *tunnel* is reached and then decreases
- Earlier work used datasets with 32×32 images (CIFAR-10, etc.) for ID training data and did not measure tunnel effect *strength* (Masarczyk et al., NeurIPS 2023)
- Their findings are contrary to widely used transfer learning approaches with ImageNet-1K backbones

Tunnel Effect Hypothesis: An *overparameterized* N - layer DNN develops two *distinct* groups:

1. *The **extractor*** consists of the first K layers, creating linearly separable representations.
2. *The **tunnel*** comprises the remaining N − K layers, compressing representations and hindering OOD generalization.

Measuring The Tunnel Effect Strength

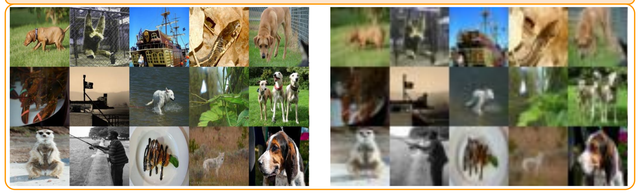


⭐ denotes the start of the tunnel

- **% OOD Performance Retained:** $r = (a_p / a_m) \times 100$, lower r indicates stronger tunnel effect
- **Pearson Correlation:** $\rho = Corr(a_{ID}, a_{OOD})$, the higher the ρ, the lower the tunnel effect will be
- **ID/OOD Alignment:** $A = (a_{ID} - c_{ID}) \times (a_{OOD} - c_{OOD})$, distinguishes low-performing and high-performing models

Here, c_{ID} and c_{OOD} denote chance accuracy. 1st two metrics use normalized accuracy whereas 3rd one uses raw accuracy

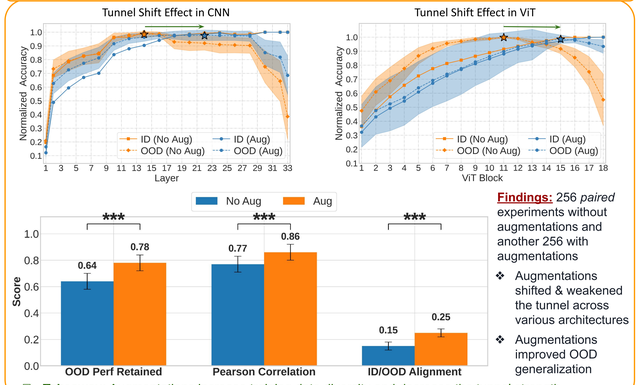
ImageNet at 224×224 VS 32×32



Experimental Setup

- Used ImageNet-100 and its variants as ID datasets and 9 widely-used OOD datasets e.g., NINCO, ImageNet-R, CIFAR-10, CIFAR-100, Oxford 102 Flowers, CUB-200, Aircrafts, Oxford Pets, and STL-10
- Studied total **8 variables:** 1) image resolution (32/64/128/224), 2) ID class count, 3) augmentations (random crop & flip), 4) overparameterization, 5) depth, 6) spatial reduction, 7) stem, and 8) CNN vs. ViT
- Trained and assessed **64 ID backbones** and **8,604 linear probes,** resulting in 512 values per metric
- Performed *paired* tests to study impact of each variable in isolation for every combination of other variables
- Paris are constructed to control for the impact of other variables
- Conducted hypothesis testing for statistical significance. *P*-values are denoted by stars according to $^*P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$, $^{****}P < 0.0001$
- Jointly analyzed and ranked variables using "SHAP Slope", our SHAP-based analysis

Research Question: How does image augmentation impact the tunnel strength?

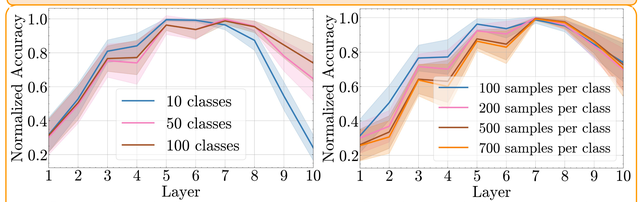


Findings: 256 *paired* experiments without augmentations and another 256 with augmentations

- Augmentations shifted & weakened the tunnel across various architectures
- Augmentations improved OOD generalization

Takeaway: Augmentations increase training data diversity and decrease the tunnel strength

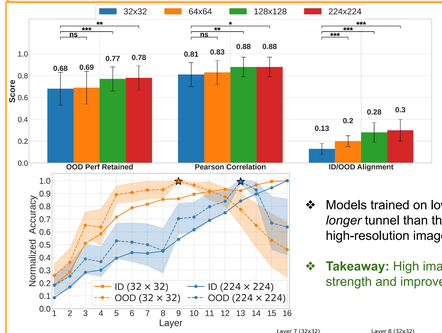
Wider Coverage of Semantics Matters More Than Data Quantity



Fixed sample size (10K) and varied class count

Fixed class count (100) and varied sample size

Research Question: How does image resolution impact the tunnel strength?



Findings: Conducted *paired* tests between models trained with 32×32 images and those trained with 64×64, 128×128, or 224×224 resolution images (48 paired experiments per group)

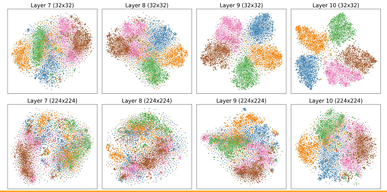
- Increasing image resolution improves OOD performance across all criteria

- Models trained on low-resolution images develop *longer* tunnel than the models trained on high-resolution images

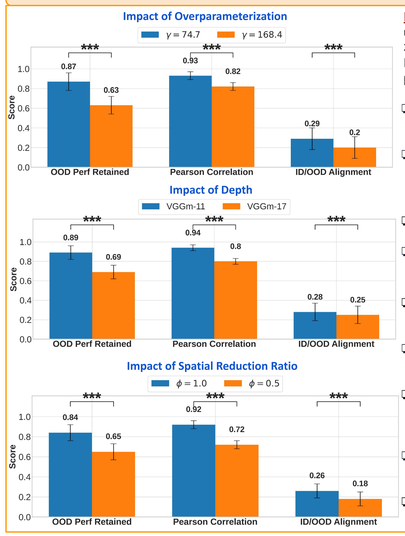
Takeaway: High image resolution reduces the tunnel strength and improves OOD generalization

Representation Compression:

- Models trained on low-resolution images exhibit much greater *representation compression* than models trained on high-resolution images
- This is likely *why* many methods that work well for low-resolution datasets e.g., CIFAR fail to perform effectively for high-resolution datasets e.g., ImageNet



Research Question: How do DNN architecture variables influence the tunnel effect?



Findings: Conducted 416 experiments using 8 DNN architectures drawn from 3 families: VGG, ResNet, and ViT. Each architecture uses the *same #* parameters across image resolutions.

- Overparameterization γ = P / N where P and N denote # DNN parameters and # training samples
- Impact of overparameterization (γ): increasing γ impairs OOD performance

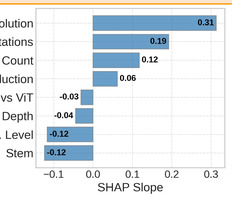
- Depth refers to number of layers for CNN and blocks for ViT
- Impact of depth: increasing depth hurts OOD generalization

- Stem refers to the kernel size (k × k) in the first layer of CNN. It is the patch size for ViT
- Impact of stem: large stem size impairs OOD generalization

- Spatial reduction ratio is defined as the ratio of the output spatial dimension to the input spatial dimension
- Impact of spatial reduction ratio (ϕ): decreasing ϕ impairs OOD transfer

- CNN vs ViT: negligible impact

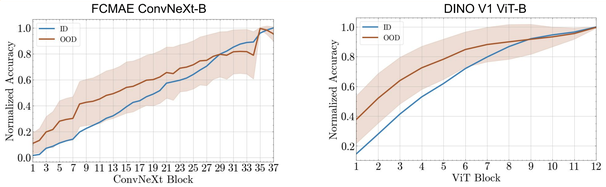
SHAP Analysis: What Variable Matters Most?



Key Findings:

- SHAP results using ID/OOD alignment as the target enable us to measure each variable's impact
- Resolution was found most dominant followed by augmentations and ID class count in terms of reducing tunnel strength
- Overparameterization level, stem, and depth increase the tunnel strength but their impact is much less than dataset variables

Do Self-Supervised Models Develop Tunnel?



Findings: Most large-scale pretrained models including self-supervised ones did not exhibit any tunnel effect

Additional Findings & Summary

Additional Findings:

- SHAP analysis using % OOD performance retained and Pearson correlation as targets revealed that
  - Dataset variables greatly improved OOD generalization where ID class count had the greatest impact followed by augmentations, spatial reduction, and resolution
  - DNN architecture variables e.g., overparameterization and depth impaired OOD generalization
  - Dataset variables had a stronger influence on the tunnel effect than DNN architecture variables
- Among 64 ID backbones, 4 did not exhibit any tunnel effect, suggesting that the tunnel effect is not a universal phenomenon and its strength depends on various factors
- Training on datasets with the properties similar to CIFAR results in very *different representations* in the "tunnel" compared to training on ImageNet, as well as much stronger tunnels
- Tunnel plays *task-specific* role and impacts forgetting in continual learning

Summary:

★ Increasing ID class count (*between-class diversity*), using augmentations (*within-class diversity*), and using higher image resolution (*hierarchical features*) greatly reduce the tunnel strength and improve OOD transfer

★ DNN variables e.g., over-parameterization and depth, increase the tunnel effect, but their impact is much smaller than that of the aforementioned dataset variables

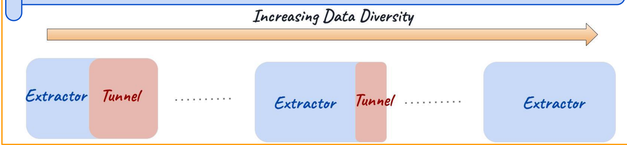
★ Concretely, we observe that increasing dataset diversity plays a major role in mitigating the tunnel effect

★ This leads us to *revise* the tunnel effect hypothesis as follows:

Revised Tunnel Effect Hypothesis: An *overparameterized* N - layer DNN develops two *distinct* groups:

1. *The **extractor*** consists of the first K layers, creating linearly separable representations.
2. *The **tunnel*** comprises the remaining N − K layers, compressing representations and hindering OOD generalization.

K is proportional to the diversity of training inputs, where if diversity is sufficiently high, N = K (no tunnel)



Acknowledgments

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